

AIM: To develop a Python program that counts even and odd numbers and calculates the sum of numbers within a given range.

### ❖ INCORRECT[ERROR] CODE:

```
print("Range Counter")

start = int(input("Enter start: "))

end = int(input("Enter end: "))

even = 0

odd = 0

sum_total = 0

while start <= end:

    sum_total = sum_total + start

    if start % 2 == 0:

        even = even + 1

    else:

        odd = odd + 1

    start = start + 1

print("Even count:", even)

print("Odd count:", odd)

print("Sum:", sum_total)

if even > odd

    print("More even numbers")

else:

    print("More odd numbers")
```

## ❖ LIST OF ERRORS:

### 1. Indentation Error in while Loop

- The statements inside the while loop were not properly indented.
- In Python, indentation is mandatory to define the block of code.

#### **Incorrect:**

```
while start <= end:  
sum_total = sum_total + start
```

#### **Correct:**

```
while start <= end:  
    sum_total = sum_total + start
```

### 2. if-else Block Outside the Loop

- The condition to check even/odd numbers was placed outside the loop due to wrong indentation.
- This prevents correct counting of even and odd numbers.

### 3. Incorrect Placement of Increment Statement

- The statement `start = start + 1` must be inside the loop.
- Otherwise, the loop may not execute properly or can result in an infinite loop.

### 4. Missing Colon (:) in if Statement

- The if condition at the end was missing a colon.

#### **Incorrect:**

```
if even > odd
```

#### **Correct:**

```
if even > odd:
```

### 5. Indentation Error in Final if-else

- The print statements inside the final if-else block were not indented.

#### **Incorrect:**

```
if even > odd:  
print("More even numbers")
```

#### **Correct:**

```
if even > odd:  
    print("More even numbers")
```

### 6. Logical Flow Issue

- Due to improper indentation and structure:
  - Even/odd count may be incorrect
  - Sum calculation may be wrong
  - Program output becomes unreliable

### Conclusion

The main errors in the program were:

- ✗ Improper indentation
- ✗ Missing colon (:)
- ✗ Incorrect block structure

**❖ CORRECT CODE:**

```
print("Range Counter")

start = int(input("Enter start: "))
end = int(input("Enter end: "))

even = 0
odd = 0
sum_total = 0

while start <= end:

    sum_total = sum_total + start

    if start % 2 == 0:

        even = even + 1

    else:

        odd = odd + 1

    start = start + 1

print("Even count:", even)
print("Odd count:", odd)
print("Sum:", sum_total)

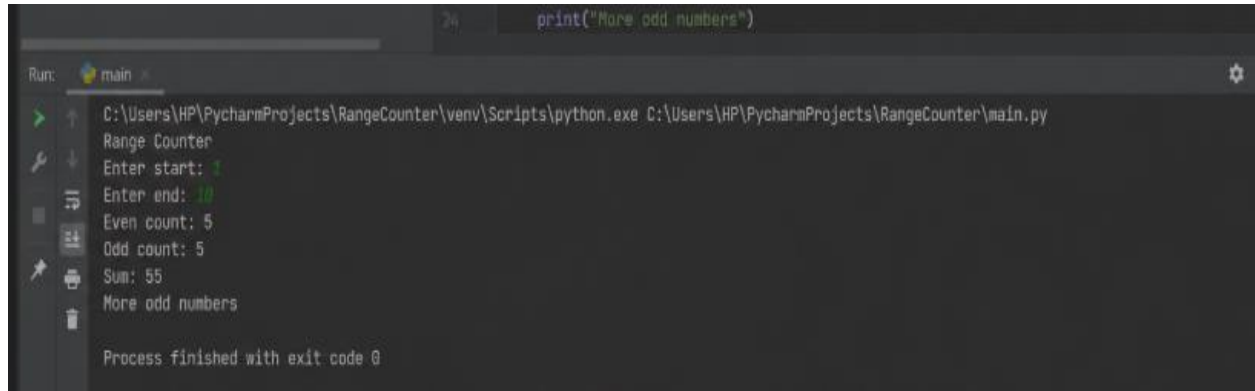
if even > odd:

    print("More even numbers")

else:
```

```
print("More odd numbers")
```

## OUTPUT OF CORRCT CODE:



```
Run: main
C:\Users\HP\PycharmProjects\RangeCounter\venv\Scripts\python.exe C:\Users\HP\PycharmProjects\RangeCounter\main.py
Range Counter
Enter start: 1
Enter end: 10
Even count: 5
Odd count: 5
Sum: 55
More odd numbers

Process finished with exit code 0
```